Module Interface Specification for Flexin'

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1 Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

2 Symbols, Abbreviations and Acronyms

See SRS Documentation at [give url —SS]
[Also add any additional symbols, abbreviations or acronyms —SS]

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3 Introduction

The following document details the Module Interface Specifications for [Fill in your project name and description —SS]

Complementary documents include the System Requirement Specifications and Module Guide. The full documentation and implementation can be found at .... [provide the url for your repo —SS]

4 Notation

[You should describe your notation. You can use what is below as a starting point. —SS]

The structure of the MIS for modules comes from **?**, with the addition that template modules have been adapted from **?**. The mathematical notation comes from Chapter 3 of **?**. For instance, the symbol := is used for a multiple assignment statement and conditional rules follow the form $(c_1 \Rightarrow r_1|c_2 \Rightarrow r_2|...|c_n \Rightarrow r_n)$.

The following table summarizes the primitive data types used by Flexin'.

Data Type	Notation	Description
character	char	a single symbol or digit
integer	$\mathbb{Z}$	a number without a fractional component in (-∞, ∞)
natural number	$\mathbb{N}$	a number without a fractional component in [1, ∞)
real	$\mathbb{R}$	any number in (-∞, ∞)

The specification of Flexin' uses some derived data types: sequences, strings, and tuples. Sequences are lists filled with elements of the same data type. Strings are sequences of characters. Tuples contain a list of values, potentially of different types. In addition, Flexin'

uses functions, which are defined by the data types of their inputs and outputs. Local functions are described by giving their type signature followed by their specification.

5 Module Decomposition

The following table is taken directly from the Module Guide document for this project.

Level 1	**Level 2**
Hardware-Hiding Module	Fitness Integration Module Persistence Module
Behaviour-Hiding Module	User and Onboarding Module Activity and Goals Module Currency Ledger Module Asset Module Transaction Order Module Portfolio Module Leaderboard and Friends Module Progress Reporting Module UI Screens Module Notification Module
Software Decision Module	Market Simulation Module Authentication and Security Module API Client Module

Table 1: Module Hierarchy

6 MIS of Fitness Integration Module

This module is M1 in the Module Guide.

6.1 Module

FitnessIntegration

6.2 Uses

[Modules this module uses —SS]

6.3 Syntax

6.3.1 Exported Constants

[Exported constants, or None —SS]

6.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

6.4 Semantics

6.4.1 State Variables

[State variables, or None —SS]

6.4.2 Environment Variables

[Environment variables, or None —SS]

6.4.3 Assumptions

[Assumptions, or None —SS]

6.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

6.4.5 Local Functions

[Local functions, or None —SS]

7 MIS of Persistence Module

This module is M2 in the Module Guide.

7.1 Module

Persistence

7.2 Uses

[Modules this module uses —SS]

7.3 Syntax

7.3.1 Exported Constants

[Exported constants, or None —SS]

7.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

7.4 Semantics

7.4.1 State Variables

[State variables, or None —SS]

7.4.2 Environment Variables

[Environment variables, or None —SS]

7.4.3 Assumptions

[Assumptions, or None —SS]

7.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

7.4.5 Local Functions

[Local functions, or None —SS]

8 MIS of User and Onboarding Module

This module is M3 in the Module Guide.

8.1 Module

User

8.2 Uses

[Modules this module uses —SS]

8.3 Syntax

8.3.1 Exported Constants

[Exported constants, or None —SS]

8.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

8.4 Semantics

8.4.1 State Variables

[State variables, or None —SS]

8.4.2 Environment Variables

[Environment variables, or None —SS]

8.4.3 Assumptions

[Assumptions, or None —SS]

8.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

8.4.5 Local Functions

[Local functions, or None —SS]

9 MIS of Activity and Goals Module

This module is M4 in the Module Guide.

9.1 Module

Activity

9.2 Uses

[Modules this module uses —SS]

9.3 Syntax

9.3.1 Exported Constants

[Exported constants, or None —SS]

9.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

9.4 Semantics

9.4.1 State Variables

[State variables, or None —SS]

9.4.2 Environment Variables

[Environment variables, or None —SS]

9.4.3 Assumptions

[Assumptions, or None —SS]

9.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

9.4.5 Local Functions

[Local functions, or None —SS]

10 MIS of Currency Ledger Module

This module is M5 in the Module Guide.

10.1 Module

CurrencyLedger

10.2 Uses

[Modules this module uses —SS]

10.3 Syntax

10.3.1 Exported Constants

[Exported constants, or None —SS]

10.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

10.4 Semantics

10.4.1 State Variables

[State variables, or None —SS]

10.4.2 Environment Variables

[Environment variables, or None —SS]

10.4.3 Assumptions

[Assumptions, or None —SS]

10.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

10.4.5 Local Functions

[Local functions, or None —SS]

11 MIS of Asset Module

This module is M6 in the Module Guide.

11.1 Module

Asset

11.2 Uses

[Modules this module uses —SS]

11.3 Syntax

11.3.1 Exported Constants

[Exported constants, or None —SS]

11.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

11.4 Semantics

11.4.1 State Variables

[State variables, or None —SS]

11.4.2 Environment Variables

[Environment variables, or None —SS]

11.4.3 Assumptions

[Assumptions, or None —SS]

11.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

11.4.5 Local Functions

[Local functions, or None —SS]

12 MIS of Transaction Order Module

This module is M7 in the Module Guide.

12.1 Module

TransactionOrder

12.2 Uses

[Modules this module uses —SS]

12.3 Syntax

12.3.1 Exported Constants

[Exported constants, or None —SS]

12.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

12.4 Semantics

12.4.1 State Variables

[State variables, or None —SS]

12.4.2 Environment Variables

[Environment variables, or None —SS]

12.4.3 Assumptions

[Assumptions, or None —SS]

12.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

12.4.5 Local Functions

[Local functions, or None —SS]

13 MIS of Portfolio Module

This module is M8 in the Module Guide.

13.1 Module

Portfolio

13.2 Uses

[Modules this module uses —SS]

13.3 Syntax

13.3.1 Exported Constants

[Exported constants, or None —SS]

13.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

13.4 Semantics

13.4.1 State Variables

[State variables, or None —SS]

13.4.2 Environment Variables

[Environment variables, or None —SS]

13.4.3 Assumptions

[Assumptions, or None —SS]

13.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

13.4.5 Local Functions

[Local functions, or None —SS]

14 MIS of Leaderboard and Friends Module

This module is M9 in the Module Guide.

14.1 Module

Leaderboard

14.2 Uses

[Modules this module uses —SS]

14.3 Syntax

14.3.1 Exported Constants

[Exported constants, or None —SS]

14.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

14.4 Semantics

14.4.1 State Variables

[State variables, or None —SS]

14.4.2 Environment Variables

[Environment variables, or None —SS]

14.4.3 Assumptions

[Assumptions, or None —SS]

14.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

14.4.5 Local Functions

[Local functions, or None —SS]

15 MIS of Progress Reporting Module

This module is M10 in the Module Guide.

15.1 Module

ProgressReport

15.2 Uses

[Modules this module uses —SS]

15.3 Syntax

15.3.1 Exported Constants

[Exported constants, or None —SS]

15.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

15.4 Semantics

15.4.1 State Variables

[State variables, or None —SS]

15.4.2 Environment Variables

[Environment variables, or None —SS]

15.4.3 Assumptions

[Assumptions, or None —SS]

15.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

15.4.5 Local Functions

[Local functions, or None —SS]

16 MIS of UI Screens Module

This module is M11 in the Module Guide.

16.1 Module

UIScreens

16.2 Uses

[Modules this module uses —SS]

16.3 Syntax

16.3.1 Exported Constants

[Exported constants, or None —SS]

16.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

16.4 Semantics

16.4.1 State Variables

[State variables, or None —SS]

16.4.2 Environment Variables

[Environment variables, or None —SS]

16.4.3 Assumptions

[Assumptions, or None —SS]

16.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

16.4.5 Local Functions

[Local functions, or None —SS]

17 MIS of Notification Module

This module is M12 in the Module Guide.

17.1 Module

Notification

17.2 Uses

[Modules this module uses —SS]

17.3 Syntax

17.3.1 Exported Constants

[Exported constants, or None —SS]

17.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

17.4 Semantics

17.4.1 State Variables

[State variables, or None —SS]

17.4.2 Environment Variables

[Environment variables, or None —SS]

17.4.3 Assumptions

[Assumptions, or None —SS]

17.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

17.4.5 Local Functions

[Local functions, or None —SS]

18 MIS of Market Simulation Module

This module is M13 in the Module Guide.

18.1 Module

MarketSimulation

18.2 Uses

[Modules this module uses —SS]

18.3 Syntax

18.3.1 Exported Constants

[Exported constants, or None —SS]

18.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

18.4 Semantics

18.4.1 State Variables

[State variables, or None —SS]

18.4.2 Environment Variables

[Environment variables, or None —SS]

18.4.3 Assumptions

[Assumptions, or None —SS]

18.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

18.4.5 Local Functions

[Local functions, or None —SS]

19 MIS of Authentication and Security Module

This module is M14 in the Module Guide.

19.1 Module

Auth

19.2 Uses

[Modules this module uses —SS]

19.3 Syntax

19.3.1 Exported Constants

[Exported constants, or None —SS]

19.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

19.4 Semantics

19.4.1 State Variables

[State variables, or None —SS]

19.4.2 Environment Variables

[Environment variables, or None —SS]

19.4.3 Assumptions

[Assumptions, or None —SS]

19.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

19.4.5 Local Functions

[Local functions, or None —SS]

20 MIS of API Client Module

This module is M15 in the Module Guide.

20.1 Module

ApiClient

20.2 Uses

[Modules this module uses —SS]

20.3 Syntax

20.3.1 Exported Constants

[Exported constants, or None —SS]

20.3.2 Exported Access Programs

Name	In	Out	Exceptions
[accessProg —SS]	-	-	-

20.4 Semantics

20.4.1 State Variables

[State variables, or None —SS]

20.4.2 Environment Variables

[Environment variables, or None —SS]

20.4.3 Assumptions

[Assumptions, or None —SS]

20.4.4 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]
- output: [if appropriate —SS]
- exception: [if appropriate —SS]

20.4.5 Local Functions

[Local functions, or None —SS]

21 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

21.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

21.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

21.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

22 Appendix — Reflection

[Not required for CAS 741 projects —SS]

The information in this section will be used to evaluate the team members on the graduate attribute of Problem Analysis and Design.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your design decisions stemmed from speaking to your client(s) or a proxy (e.g. your peers, stakeholders, potential users)? For those that were not, why, and where did they come from?
4. While creating the design doc, what parts of your other documents (e.g. requirements, hazard analysis, etc), it any, needed to be changed, and why?

5. What are the limitations of your solution? Put another way, given unlimited resources, what could you do to make the project better? (LO_ProbSolutions)

6. Give a brief overview of other design solutions you considered. What are the benefits and tradeoffs of those other designs compared with the chosen design? From all the potential options, why did you select the documented design? (LO_Explores)

7. (After you have implemented another team's module, which means this isn't filled in until after the original deadline). What did you learn by implementing another team's module? Were all the details you needed in the documentation, or did you need to make assumptions, or ask the other team questions? If your team also had another team implement one of your modules, what was this experience like? Are there things in your documentation you could have changed to make the process go more smoothly for when an "outsider" completes some of the implementation?

   Your reflection should be about concepts learned in your SE education, such as modular decomposition, cohesion, coupling, information hiding, etc. You should critique the quality and utility of the specification you received.

   You should create a PR for your implemented module. A link to the PR should be part of your reflection.